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PUNE TRAFFIC CONGESTION: REALITY, CAUSE AND REGULATION A CASE STUDY

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ABSTRACT

Traffic Congestion or traffic jams is one of the major issues in most metropolitan cities like Pune. As we know Pune city is a well-developed and popularly known for Oxford for east for its education many students come from different cities secondly many young people come for jobs because of booming IT companies. Due to this city has become densely populated. Today the Pune's population is around 35 lakhs. Registered vehicles are 36.2 lakh which is exceeding the human population. During the peak hours i.e. 09:00 to 11:00 in the morning and 6:00 to 8:00 in the evening traffic situation gets worst and chaotic where most of the schools and colleges are located especially in Peth areas having narrow roads and lane, IT offices located in Hinjewadi and Kharadi and Industrial Zone at Bhosari but nowadays you can find anytime of the day as well. Some of the issues which create the traffic problems there are no proper planning, violation of the rules & regulation, bad construction. Because of improper management there is possibility of roads accidents and traffic congestions. The main cause of traffic jams in Pune is road side parking which should be banned and penalized. Traffic jams can be improvised on various strategies such as road broadening, improved road infrastructures, penalty to the traffic law breakers and application of Fly over, properly planned traffic junctions, no hawkers zone etc. To lessen the traffic problems there should be effective traffic management system and proper implication of traffic rules and system. ICT will make real transformation in Traffic Congest.

Keywords: Traffic Jams, Conjunction, Road Broadening, ICT

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1. INTRODUCTION

As per Pune Traffic Office, the total number of registered vehicles in Pune is currently at 3.62 million (36.2 lakhs) which is more than currently human population at 3.5 million (35 lakh). This makes Pune the first city in India to which registered vehicles exceed human population.

This consensus was conducted by the RTO on 31st March 2018 with an increase of over 2.80 lakh vehicles taking previous year's figures from 33.37 lakhs (2016-17) to 36.27 lakhs (2017-18). Giving a segment-wise breakdown of all vehicles registered with the RTO Pune over the past year reveals that 6,45,683 four wheelers, 2,73,147 two wheelers, 53,225 auto rickshaws and 28,344 taxis were registered. It was also 38,598 vehicles in the truck, bus and tempo category that were registered.

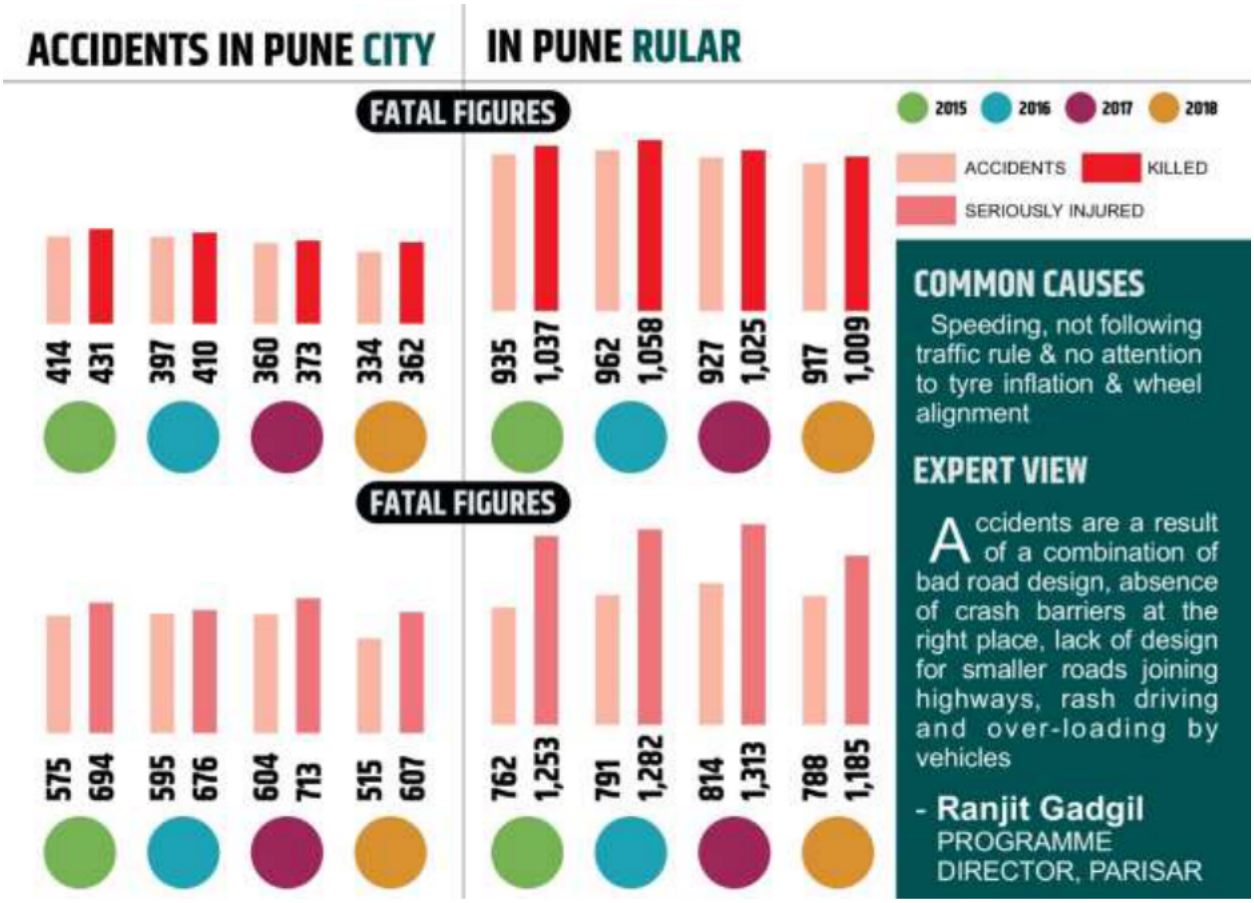

Figure 1 Statistics of Accident in Pune; Source: Published data by authorities

Citizens are not using the public transport facility which is highly required to utilized and less use of private vehicles which leads to save money as well as getting solution of traffic jam. Footpath sealer and the shop owners are also leading to traffic jam problem. Pune police and Pune Municipal corporation need to work together to resolve the issues related to encroachment which will reduce the traffic conjunction issues. Cops have to work on the four encouragement category are which are as 1) encouragement is totally illegal 2) shop owner have a license but has given on rent 3) have a license but used additional space as extension which disturbs the public transport 4) visiting a shop creates a traffic jam. Municipal corporation need to focus on basic need of infrastructure else every passing day increases the traffic problem very fast. The fatality rate is one person a day or 10 to 15 a week which is very high because of unorganized, undisciplined traffic.

2. WRONGLY BUILT FLYOVERS AND TRAFFIC CONGESTION

It has been observed in recent survey that Pune is becoming the fastest growing city in India. In Smart City project, the Pune Municipal Corporation (PMC) claim that that the Pune is aspires to become global urban centre in India.

In the footnote PMC says that Pune citizen enjoy safe and livable environment with good connectivity. In the Environment Status Report (ESR) it suggests that the city is witnessing alarming rise in the levels of air and noise pollution.

TOTAL NUMBER OF VEHICLES IN DISTRICT

Division	Number of Vehicles
Pune (MH 12)	38,88,690
Pimpri Chinchwad (MH 14)	18,75,821
Baramati (MH 42)	4,06,085
Total	**61,70,596**

NEW VEHICLES REGISTERED

Division	2016 -17	2017 -18	2018 -19
Pune (MH 12)	2,70,307	2,91,117	2,61,410
Pimpri Chinchwad (MH 14)	1,25,377	1,54,591	1,54,444
Baramati (MH 42)	23,665	32,681	31,530

NUMBER OF VEHICLES IN PUNE

2014-15	28,50,451
2015-16	30,72,003
2016-17	33,37,370
2017-18	36,27,280
2018-19	38,88,690

VEHICLES REGISYERED IN 2018-19 IN PUNE

Two-wheeler	1,76,314
Four-wheeler	47,617
Autorickshaw	16,044
Cab-Taxi	6,732
Other	14,703
Total	**2,61,410**

Figure 2 Statistics of Vehicle sold i in Pune; Source: Published data in News Paper

Pune is the highly vehicle density city one of the survey claim that there are 753 vehicles per 1,000 people in the city. And it is difficult that fast traffic move on city's roads is difficult because of physical limitations of the city infrastructure. Because of 34 villages with around 450 square kilometers of area from surrounding, Pune become one of the largest cities in India. The joint population of Pune Municipal Corporation and Pimpri-Chinchwad Muncipal Corporation has already cross 55 lakh.

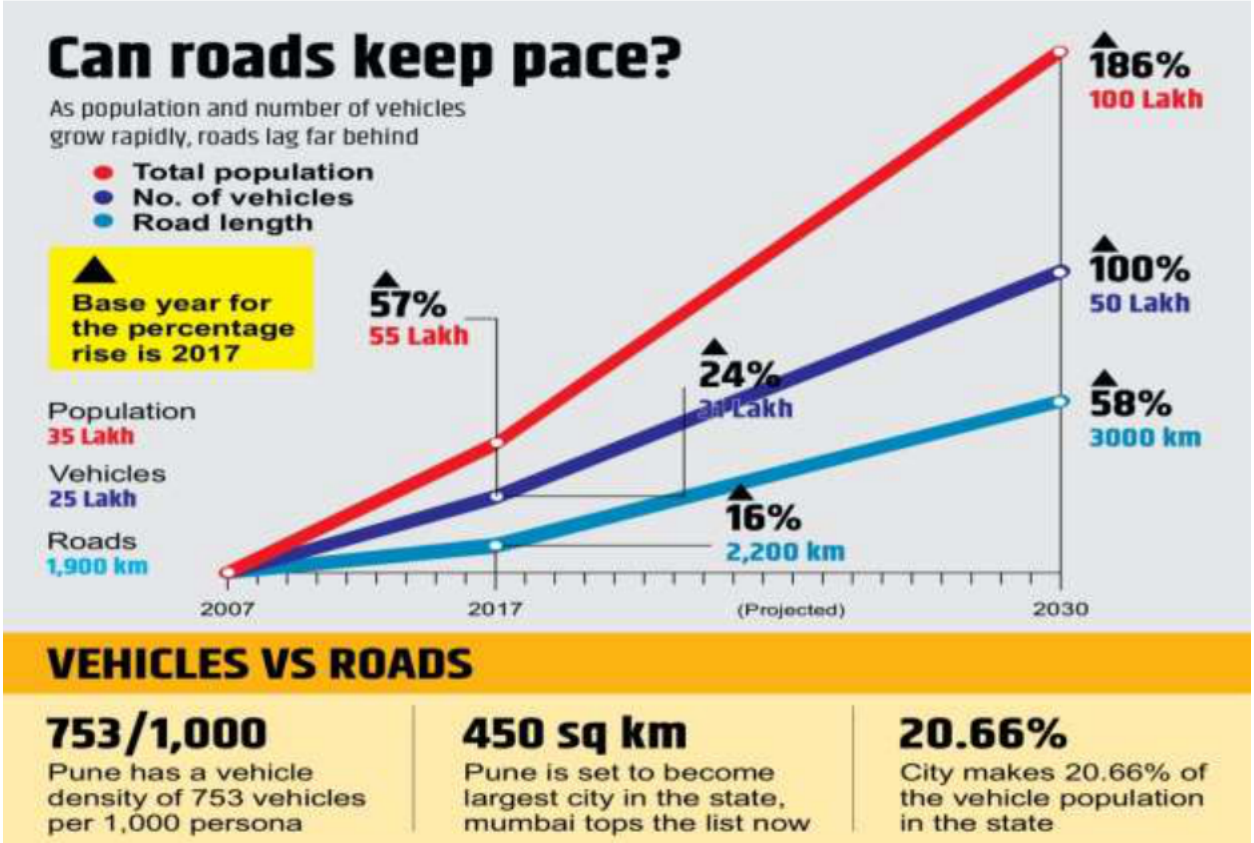

Figure 3 Statistics of Vehicles on Pune; Source: Published data by Municipal Corporation

As per the population size Pune is 8th rank city and fastest growing city in India. In term of GDP value Pune rank 5th with Rs. 1,065 per person. In Pune there are 811 colleges and educational institution and 9 state and private universities which is second largest in India. Under central government scheme in last 10 years Pune spend Rs. 30 crore. Usually city locations where roads or infrastructure is under-construction are most jam-packed. Even areas near commercial zones of office areas are jammed with traffic especially during working hours when everybody are in rush to reach office on time. Mainly Satara Road, Karve Road, Pune-Nagar Road, Ganeshkhind, Swargate, Hinjawadi are facing major problem of daily traffic jam. Pune Municipal Corporation planning to start water transport scheme on Mutha river to avoid the traffic conjestion.

The city's traffic will be improved only if there is strong infrastructure in place which is currently lacking in Pune Municipal Corporation only build around 40 flyovers to avoid the minimum traffic jam in past few years. Wrong design of flyovers created more traffic problems in the various locations in Pune city. There is lack of coordination between different government agencies and corporation. Pune public is facing problem where one government body constructs new roads and another body carries out digging. Pune Metropolitan Region Development Authority (PMRDA) has now planned that outer road to decongest the city. There is planning of proposed 129-kilometer long ring road will use to take load of heavy vehicles. Pune citizens expect good infrastructure projects along with better traffic policing to avoid traffic jams, which has become an everyday scene in the city.

3. CAUSES OF TRAFFIC CONGESTION

Narrow roads in Pune are not much widespread, have small roads and lanes. Due to illegal possession these small roads are getting narrower and becoming reason to traffic jam. Traffic signals are out of sync many times on purpose or occasionally when the computers are malfunctioning ending creating traffic chaos in the city. Increase in population will lead to increase in vehicle which is a bad traffic management and this could be vital reason for traffic congestion. Illegal parking in No-parking zone area, parking and halting including auto-riskshaws, cars on the busy streets, people don't obey the parking policy and violators are not even penalized. Due to high income purchasing power of the public has increased in each house almost each person is having vehicle. Existing roads and highway are not supportive or changing according to number of increase vehicles. As a result vehicle congestion is increasing at alarming rate. Obstacles in the road causing a blockage and merger. These can be any of the following:

- Double parking
- Road work
- Lane closure due to utility work
- Road narrowing down
- An accident

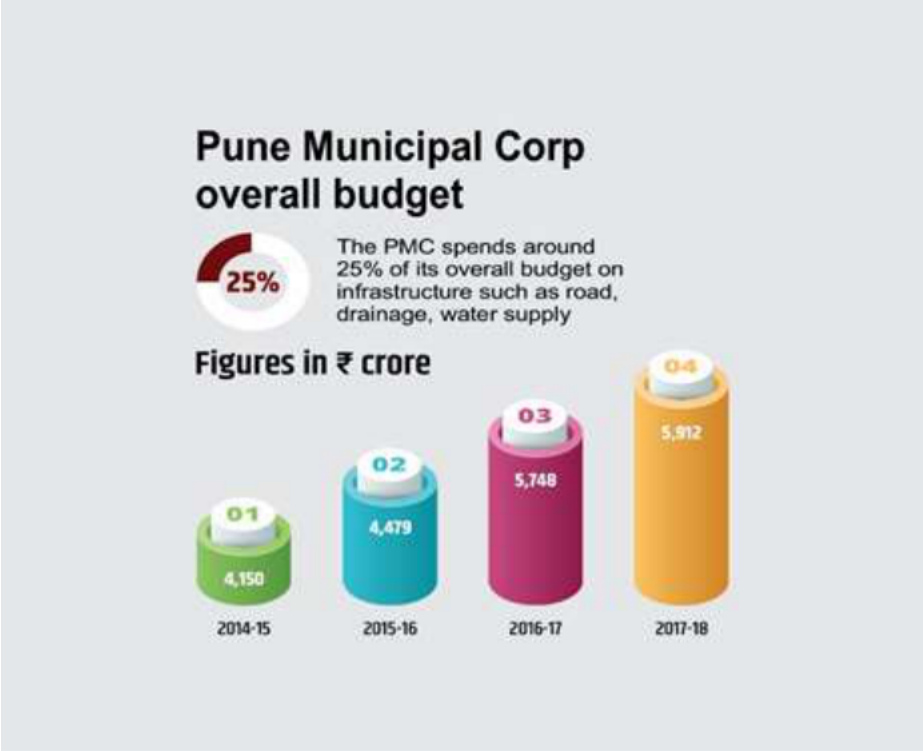

Figure 4 Statistics of Pune Municipal Corporation

Due to adequacy of space people have started parking on the footpath, drives on footpath, wrong side driving which causes difficult for pedestrian while walking. Corporation or traffic control department are not creating awareness of traffic rules among the people and educate them to follow the rules. There is no difference in highly qualified professional like engineers and doctor's behavior and illiterates on the roads. Nobody follows the traffic norms which ultimately lead to traffic jam. Heavy vehicles transit leads to increases in traffic jams in city road and responds to accidents and incidents which also results into congestion. Not following traffic signals like skipping the signals are common practices of citizen drivers. Wrong side driving, overtaking, taking a short cut through small lanes mainly cars, autos or any four or three-wheel vehicle, merging in lane, etc. are common regions for the traffic congestions in traffic of city. Insufficient public transportation encourages people to use their own vehicles that contribute in increasing volume of vehicles on the roads. Fast development of city due to employment and education has pulled huge population from across the nation which leads to increase in traffic. Traffic congestion contributes increase in vehicle emissions and degrades ambient air quality causing serious pollution problems like carbon monoxide and smog. Traffic fumes contain harmful chemicals that pollutes the atmosphere and these chemicals are harmful to human's health. Road traffic emissions produces gases contribute to the global warming.

4. RESEARCH METHODOLOGY

The research is depending on (N=225) sample size and primary data has been collected through online as well as offline questionnaire using personal and student contact in Pune city. Data is collected through various cluster of Pune city depending on traffic density as a priority. Interval scale and like scale has been use for data collection and non-parametric tests are used for analysis. The research is cause and effect relationship in the research.

4.1. Suggestions to help curb the traffic problem in Pune

- More traffic police or officers for controlling traffic should be appointed on the roads/signals as they are only one who direct the traffic or serve the road traffic policing unit enforcing the rules on the roads.
- High tech solutions like GPRS, CCTVs, sensors and other devices need to install in order to provide 24/7 traffic advice. This covers different types of incidents one can expect to face on the roads like congestion, weather conditions, road work and many others things. All the messages sent to commuters on the phone in advance to prevent traffic jams.
- Educating people by providing information, handouts and displays to encourage safe driving and usage of the roads.

2 lakh vehicles are added to the city's roads every year

70% of car buyers in Pune prefer the petrol variant

70% Two - wheelers constitute

MORE THAN DOUBLE IN A DECADE

YEAR	REGISTERED VEHICLES
• 2014	26.48
• 2011	20.87
• 2010	19.03
• 2007	15.04
• 2005	12.24

ANNUAL ADDITION

YEAR	BIKES	CARS
• 2015	21.52	5.00
• 2014	19.99	4.66
• 2013	18.50	4.20

	Pollutant	Sunday	Monday	After 2 Days
PUNE	PM 10 (µg/m3)	124.6	105.9	106.16
	PM 2.5 (µg/m3)	66.3	59.5	59.1
DELHI	PM 10 (µg/m3)	359.4	356.4	357.19
	PM 2.5 (µg/m3)	225.8	223.4	216.6
MUMBAI	PM 10 (µg/m3)	149.3	147.7	147.57
	PM 2.5 (µg/m3)	110.5	110.2	108.1

CONTRACT	Sector	PM 10	PM 2.5	Black carbon
	• Industry	3.29	6.34	10.87
	• Transport	35.71	35.63	34
	• Biofuel	1.47	2.66	1.99
	• Others	5.66	4.03	Nil

- Major problem can be resolve by training people the traffic disciplines like lane discipline, overtaking rules, speed limits, stopping distances, low & high beams of light, zebra crossing meant for pedestrians, how to deal with roundabouts, showing courtesy on roads, red light discipline and free turns, blind spots and other symbols on roads, etc.
- Encouraging people to use effective, frequent, fast and quality public transport can help in mass traveling and reducing number of vehicles on roads for commuting purpose.
- Proper and sufficient parking infrastructure should be provided in the highly population density area. Malls, multiplexes and corporate offices should provide parking to its customers and employees in their premises.
- Road widening, removing encroachments and building flyovers, signal less circulars, subways are some of the long term solutions that resolve some of problems in the area.
- The roads are dug up for various reasons like to install utility lines-sewage, water pipelines, cooking gas connection, electricity and optical fiber network and telephone. Timing of digging should be fixed and should be on fast track so that the traffic does not get disrupt. There should be deadline to fix the dug roads. If the civic agencies do not meet the deadline then then should be penalized.

- Illegal hoarding should be curbed to ensure the safety of road commuters. Roads are dug up for installing pandal during festive seasons, thereafter the potholes are not filled and patches are not repair. Strict action should be taken for non-repairing the roads. Folding method should be used.
- Carpooling is good option to reduce to contribution to traffic roads and public transport is also one of the solutions to reduce the traffic on roads. Working class people should use this method of carpooling during the peak hours while going and coming to office. This will also help in reducing the maintenance costs of the vehicle users while reducing the amount of pollutants vehicle emitted to the environment.
- No of people from the family traveling on should avoid commuting and use appropriate vehicle Two-wheeler should be preferable as and when possible.
- Air pollution created by traffic congestion can be reduced by using traffic control powers by the local authorities. They need to monitor the level of traffic on their roads and set traffic reduction targets.
- The footpaths should be cleared, free from illegal parking, illegal driving, small shops and hawkers which occupy half the space of the footpaths causing hindrance to pedestrian. Regular monitoring of administration by local authorities will prevent such occupancy and also traffic officers should penalize such drivers using shortcuts on footpath.
- Construct short fly over at the meeting road for all the age's pedestrian to lighting the congestion.

4.2. Road Safety Tips

Following are the majorly key factors affecting on the road safety

- High requirement of Strengthening of public transport system is required.
- Dedicated parking and non-parking zones should be assign and operated.
- Enforcement and Education need to provide in the various area of operation.
- One-way traffic movement on streets should be assign and improve
- Hawkers zones should not assign in the area of highly travel density area.
- There should be separate and dedicated exclusive bus-lanes should be assign for public transport.
- Construction of flyovers, subways, sky walks and cross over the road should be increase
- Restrictions on turning of vehicle on the road side need to improve
- Tidal flow operations.
- Side streets should be closed for public use.
- There should be Proper plan of traffic junctions on the road.
- New mode of fast transport need to introduce and implement in the city.
- Don't use mobile while driving on road
- Use of seat belt and helmet while driving
- Don't drink and drive on road
- Slow down the speed as and when required
- Take the appropriate brake while driving

- Use of child belt while driving
- Keep the safe distance while driving

5. CONCLUSION

The main cause of traffic jams in Pune is road side parking which should be banned and penalized. Traffic jams can be improvised on various strategies such as road broadening, improved road infrastructures, penalty to the traffic law breakers and application of Fly over, properly planned traffic junctions, no hawkers zone etc. To lessen the traffic problems there should be effective traffic management system and proper implication of traffic rules and system. ICT will make real transformation in Traffic Congest.

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